

Majorization and Uhlmann's Theorem

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The two slides headed *Mathematical tools* from *Complexity of Mutual Simulation of Hamiltonian Dynamics*, Ludwig-Maximilians-Universität München, 12 July 2002. They are set here on their own because they are the earliest record I have of my working with majorization, which came back much later in my work on spectral graph theory. The wording and the notation are those of the 2002 slides; only the typesetting is new.

Vector majorization

Let $x = (x_1, x_2, \dots, x_d) \in \mathbb{R}^d$.

Define $x^\downarrow = (x_1^\downarrow, x_2^\downarrow, \dots, x_d^\downarrow) \in \mathbb{R}^d$ as the vector whose components are the same as those of x , but ordered so that

$$x_1^\downarrow \geq x_2^\downarrow \geq \dots \geq x_d^\downarrow.$$

x majorizes y , written $x \succeq y$, if

$$\sum_{j=1}^k x_j^\downarrow \leq \sum_{j=1}^k y_j^\downarrow$$

for $k = 1, \dots, d-1$ and

$$\sum_{j=1}^d x_j^\downarrow = \sum_{j=1}^d y_j^\downarrow.$$

Operator majorization

Let A and B be two Hermitian matrices of size d .

We say A majorizes B , written $A \succeq B$, if

$$\lambda(A) \succeq \lambda(B),$$

where $\lambda(A)$ and $\lambda(B)$ are the spectra (the vectors of eigenvalues) of A and B , respectively.

Uhlmann's theorem

We have $A \succeq B$ if and only if

$$A = \sum_{j=1}^N p_j U_j^\dagger B U_j,$$

where $\{p_j\}$ is a probability distribution and $\{U_j\}$ is a set of unitaries.